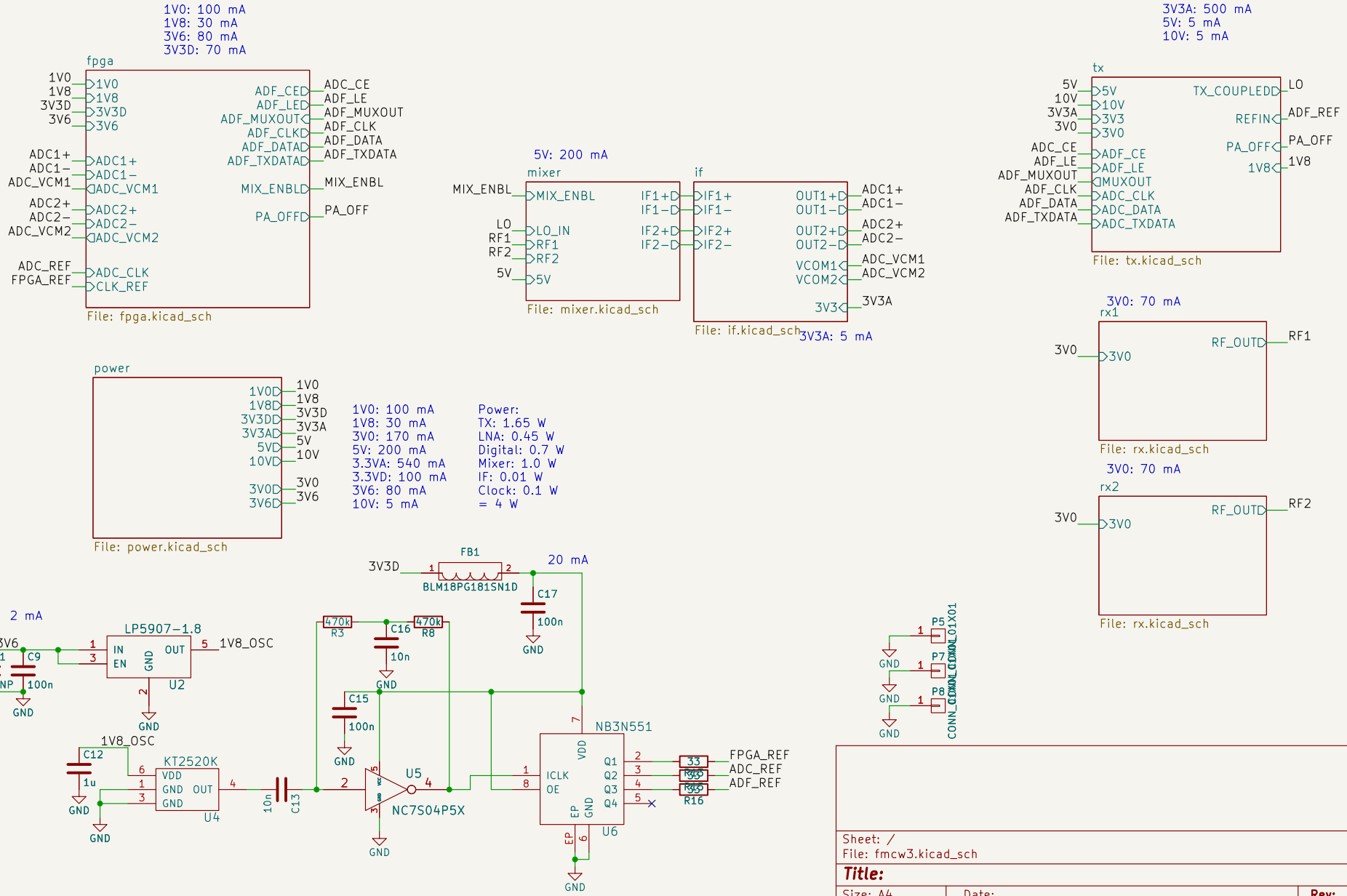
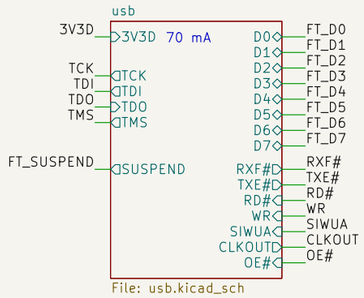
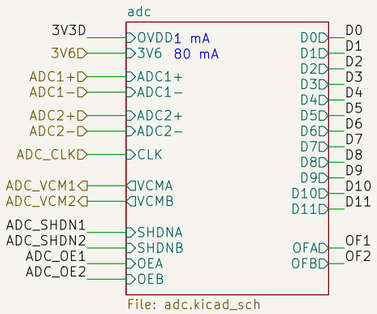
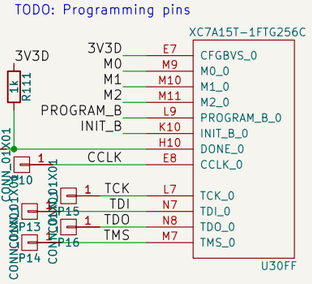
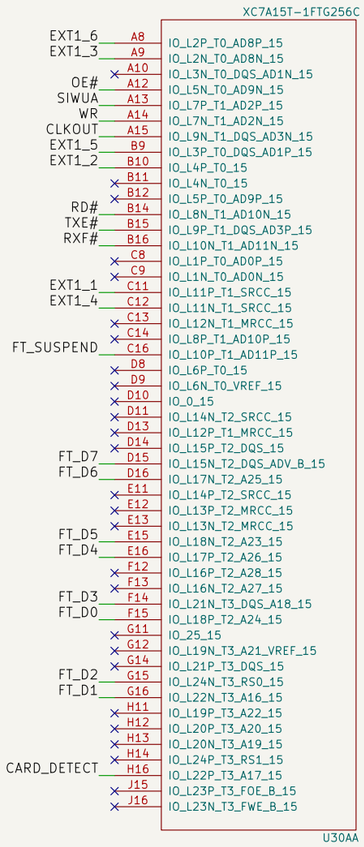
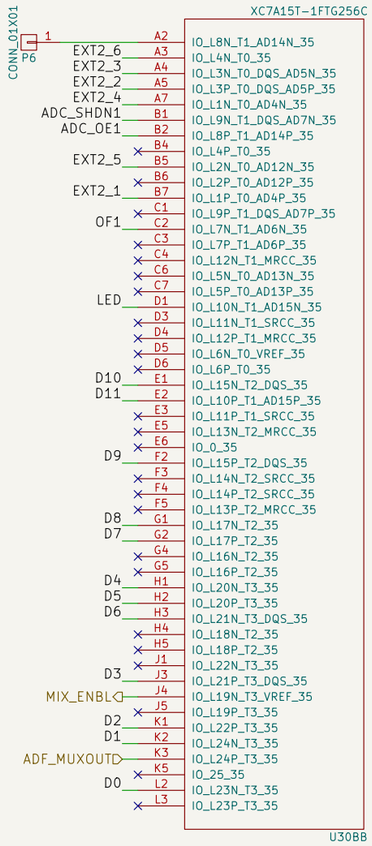

1V0: 100 mA
1V8: 30 mA
3V6: 80 mA
3V3D: 70 mA
fpga
1V0
1V8
3V3D
3V6
ADC1+
ADC1−
ADC_VCM1
ADC2+
ADC2−
ADC_VCM2
ADC_REF
FPGA_REF
ADC_CE
ADF_LE
ADF_MUXOUT
ADF_CLK
ADF_DATA
ADF_TXDATA
MIX_ENBL
PA_OFF
File: fpga.kicad_sch
5V: 200 mA
mixer
LO
RF1
RF2
5V
IF1+
IF1−
IF2+
IF2−
File: mixer.kicad_sch
if
OUT1+
OUT1−
OUT2+
OUT2−
VCOM1
VCOM2
3V3
3V3A: 5 mA
File: if.kicad_sch
3V0: 27 mA
3V3A: 500 mA
5V: 5 mA
10V: 5 mA
tx
5V
10V
3V3A
3V0
TX_COUPLED
REFIN
PA_OFF
1V8
File: tx.kicad_sch
3V0: 70 mA
rx1
RF_OUT
RF1
File: rx.kicad_sch
rx2
RF2
power
3V3D
3V3A
5V
10V
1V0: 100 mA
1V8: 30 mA
3V0: 170 mA
5V: 200 mA
3.3VA: 540 mA
3.3VD: 100 mA
3V6: 80 mA
10V: 5 mA
Power:
TX: 1.65 W
LNA: 0.45 W
Digital: 0.7 W
Mixer: 1.0 W
IF: 0.01 W
Clock: 0.1 W
= 4 W
File: power.kicad_sch
FB1
BLM18PG181SN1D
20 mA
C17
100n
2 mA
LP5907−1.8
IN
EN
GND
OUT
1V8_OSC
U2
C1
C9
DNP
470k
R3
R8
C16
10n
C15
KT2520K
C12
1u
VDD
U4
C13
U5
NC7S04P5X
NB3N551
ICLK
OE
Q1
Q2
Q3
Q4
EP
U6
33
R16
FPGA_REF
ADC_REF
ADF_REF
P5
P7
P8
CONN_01X01
Sheet: /
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 1/10

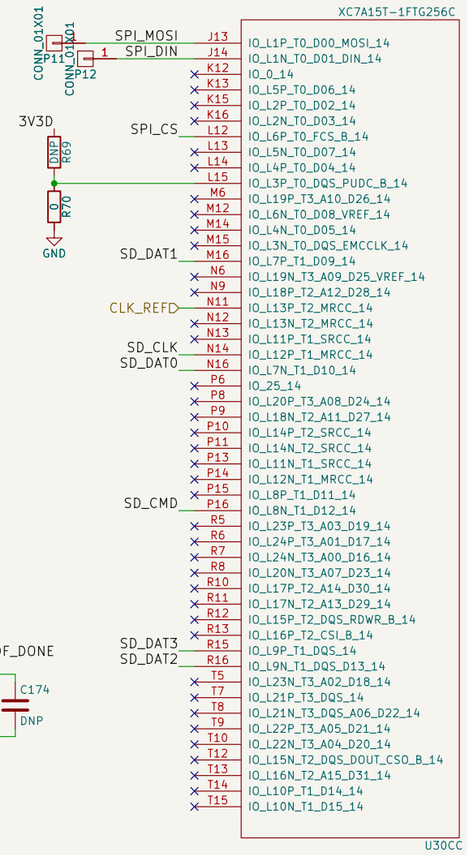


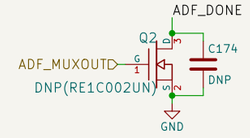








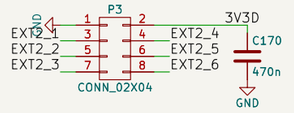


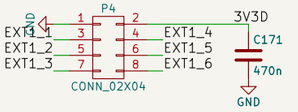


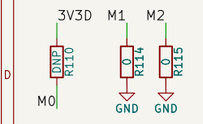
Power supply ramp time: 0.2 ms – 50 ms

Quiescent current
1V0: 97 mA
1V8: 22 mA
3V3: 2 mA

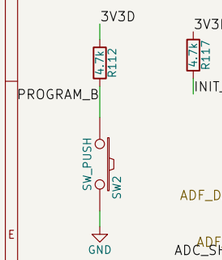
Power−on current
1V0: Iq + 180 mA
1V8: Iq + 40 mA
3V3: Iq + 40 mA

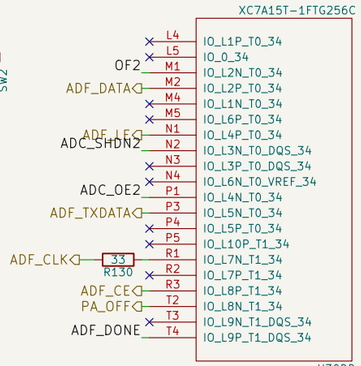


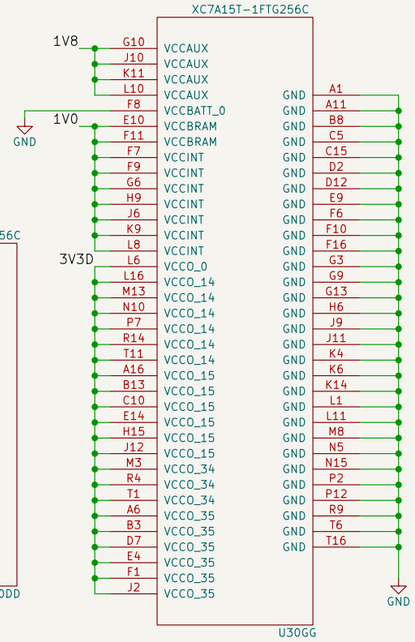


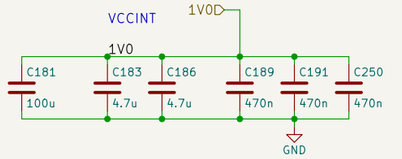


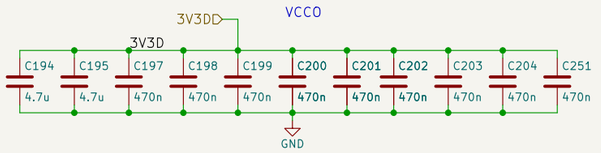
Mode M[2:0]
Master serial 000
Master SPI 001
Master BPI 010
Master SelectMap 100
JTAG 101
Slave SelectMap 110
Slave serial 111

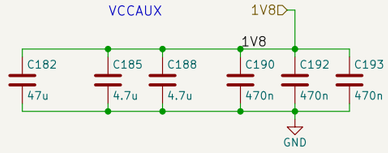




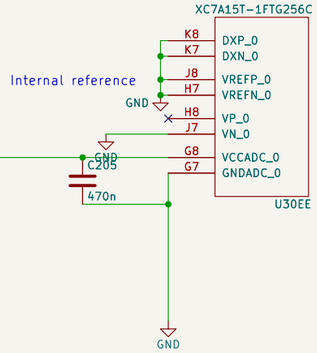


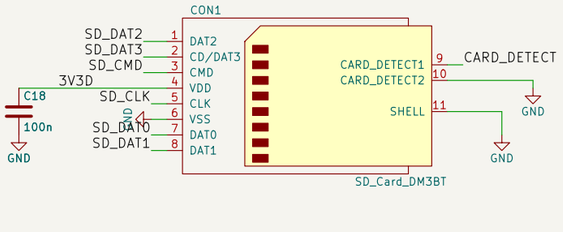




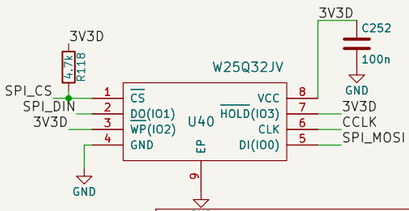


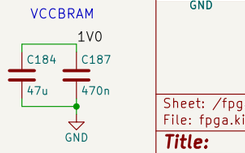




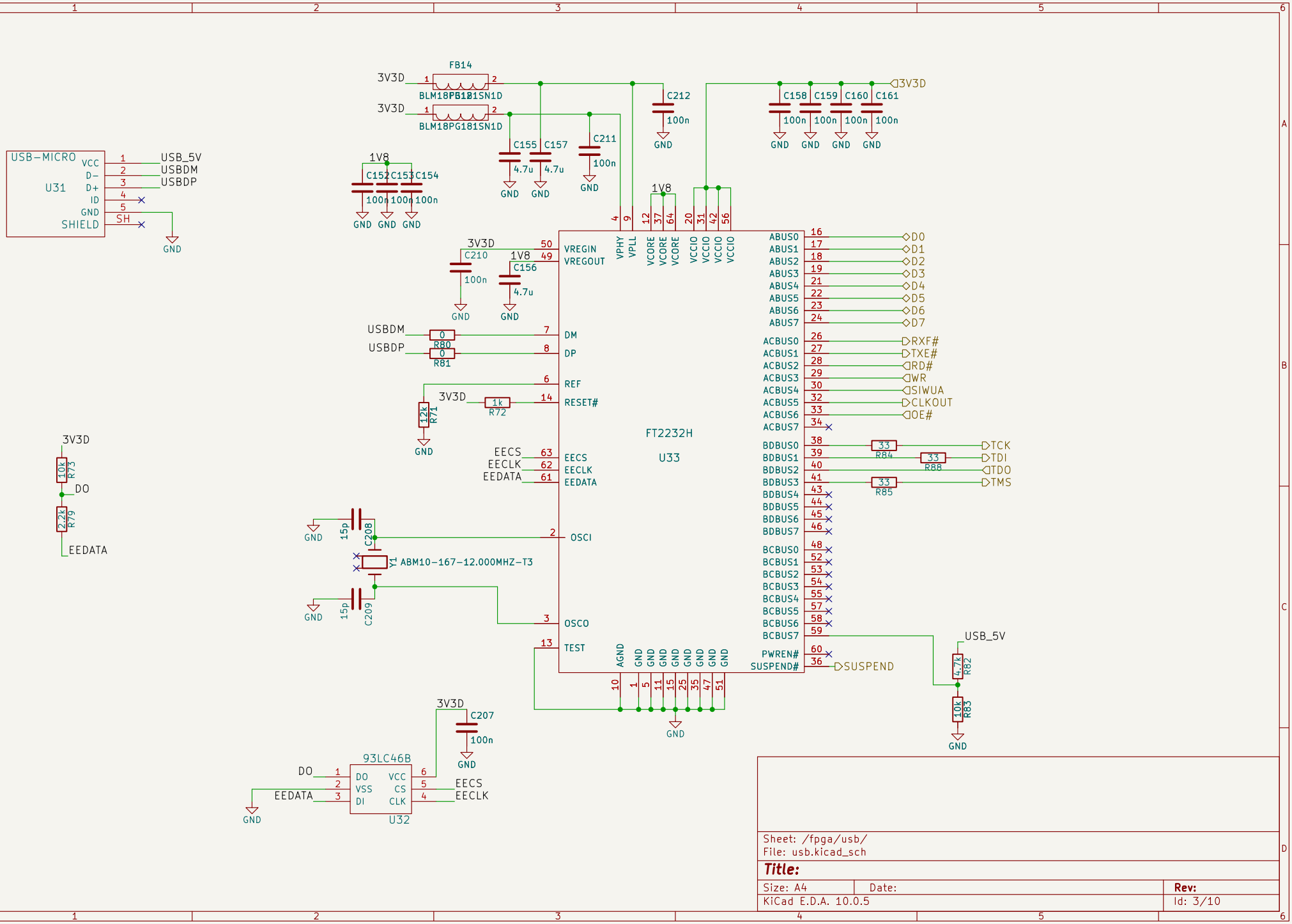


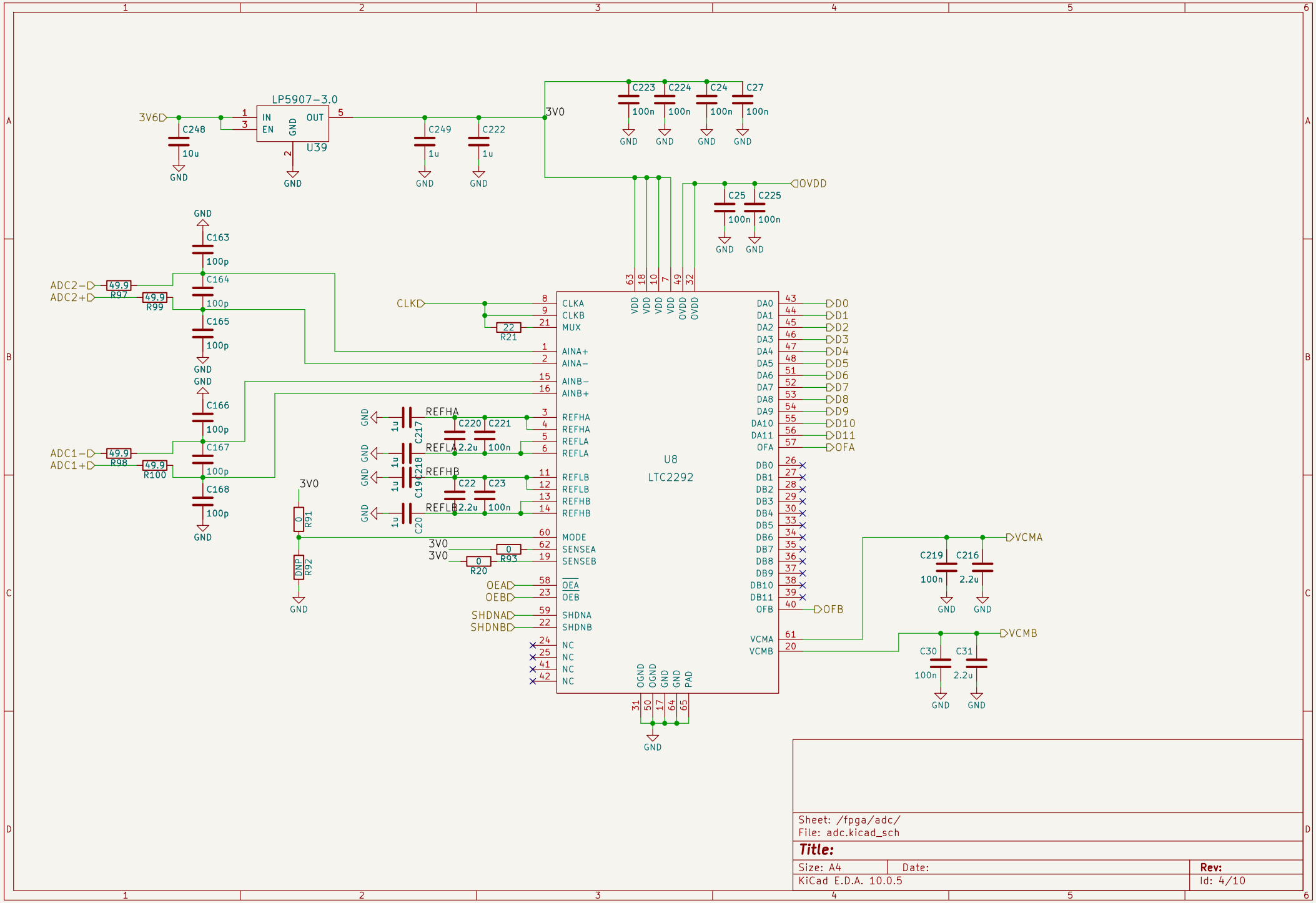




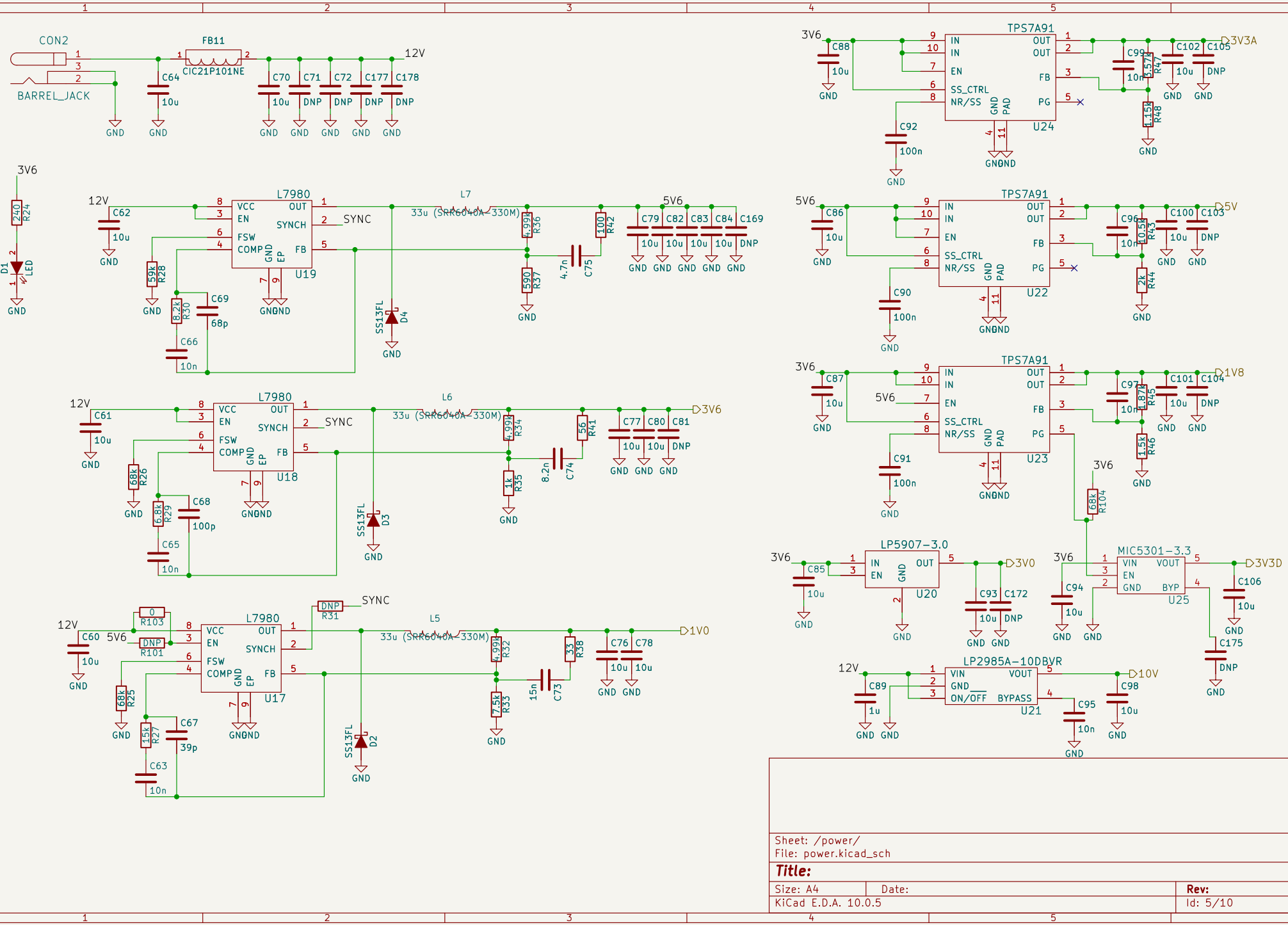


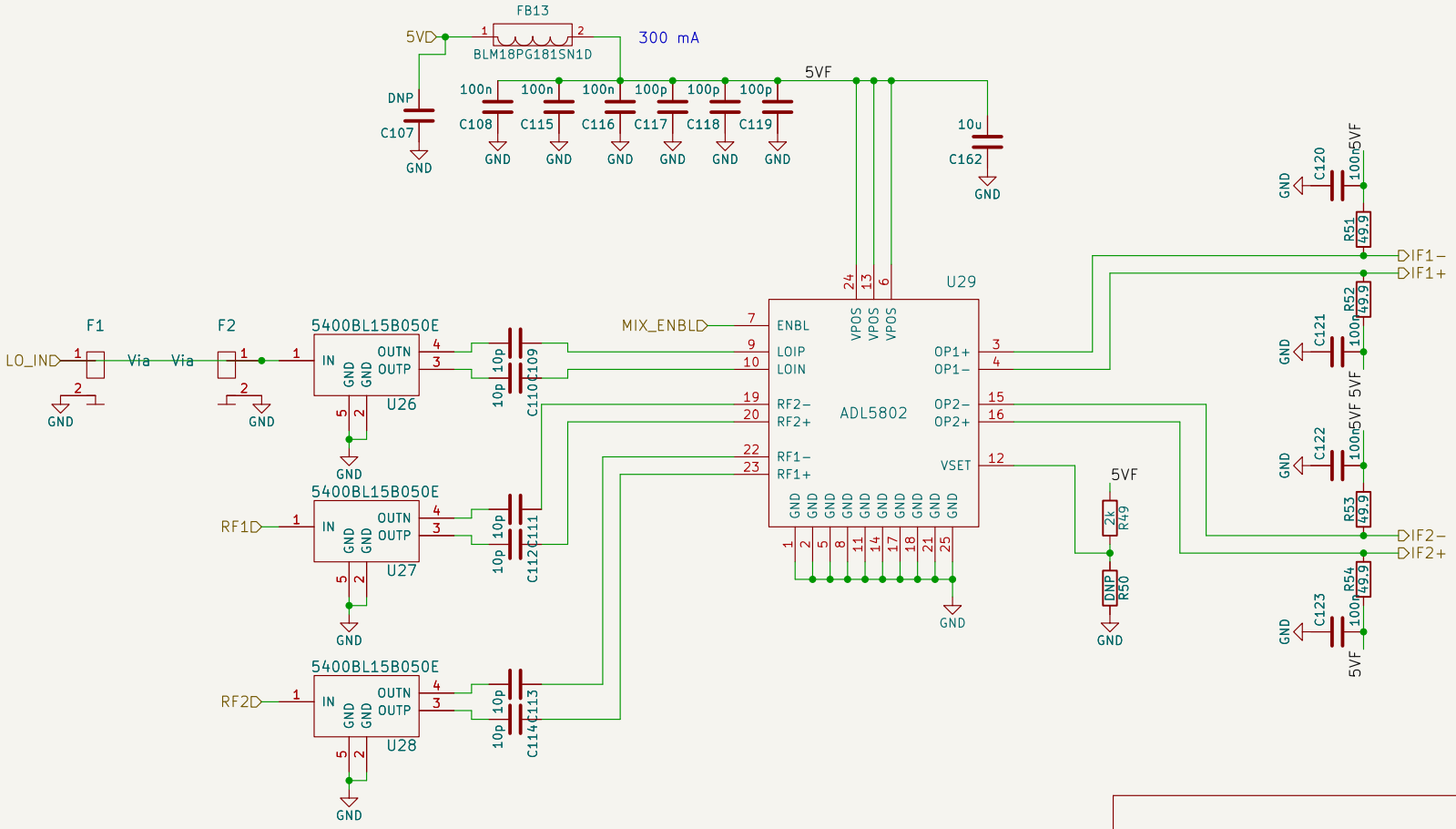
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 2/10


USB-MICRO
U31
VCC
D-
D+
ID
GND
SHIELD
USB_5V
USBDM
USBDP
FB14
BLM18PG121SN1D
3V3D
1V8
C152
C153
C154
C155
C157
C211
C212
C158
C159
C160
C161
100n
4.7u
GND
C210
C156
VREGIN
VREGOUT
VPHY
VPLL
VCORE
VCCIO
USBDM
USBDP
R80
R81
DM
DP
REF
RESET#
R71
R72
1k
12k
EECS
EECLK
EEDATA
FT2232H
U33
ABUS0
ABUS1
ABUS2
ABUS3
ABUS4
ABUS5
ABUS6
ABUS7
D0
D1
D2
D3
D4
D5
D6
D7
ACBUS0
ACBUS1
ACBUS2
ACBUS3
ACBUS4
ACBUS5
ACBUS6
ACBUS7
RXF#
TXE#
RD#
WR
SIWUA
CLKOUT
OE#
BDBUS0
BDBUS1
BDBUS2
BDBUS3
BDBUS4
BDBUS5
BDBUS6
BDBUS7
TCK
TDI
TDO
TMS
R84
R88
R85
33
BCBUS0
BCBUS1
BCBUS2
BCBUS3
BCBUS4
BCBUS5
BCBUS6
BCBUS7
PWREN#
SUSPEND#
SUSPEND
OSCI
OSCO
TEST
AGND
C208
C209
15p
Y1 ABM10-167-12.000MHZ-T3
R73
R79
10k
2.2k
DO
93LC46B
U32
C207
R82
R83
4.7k
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Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 3/10

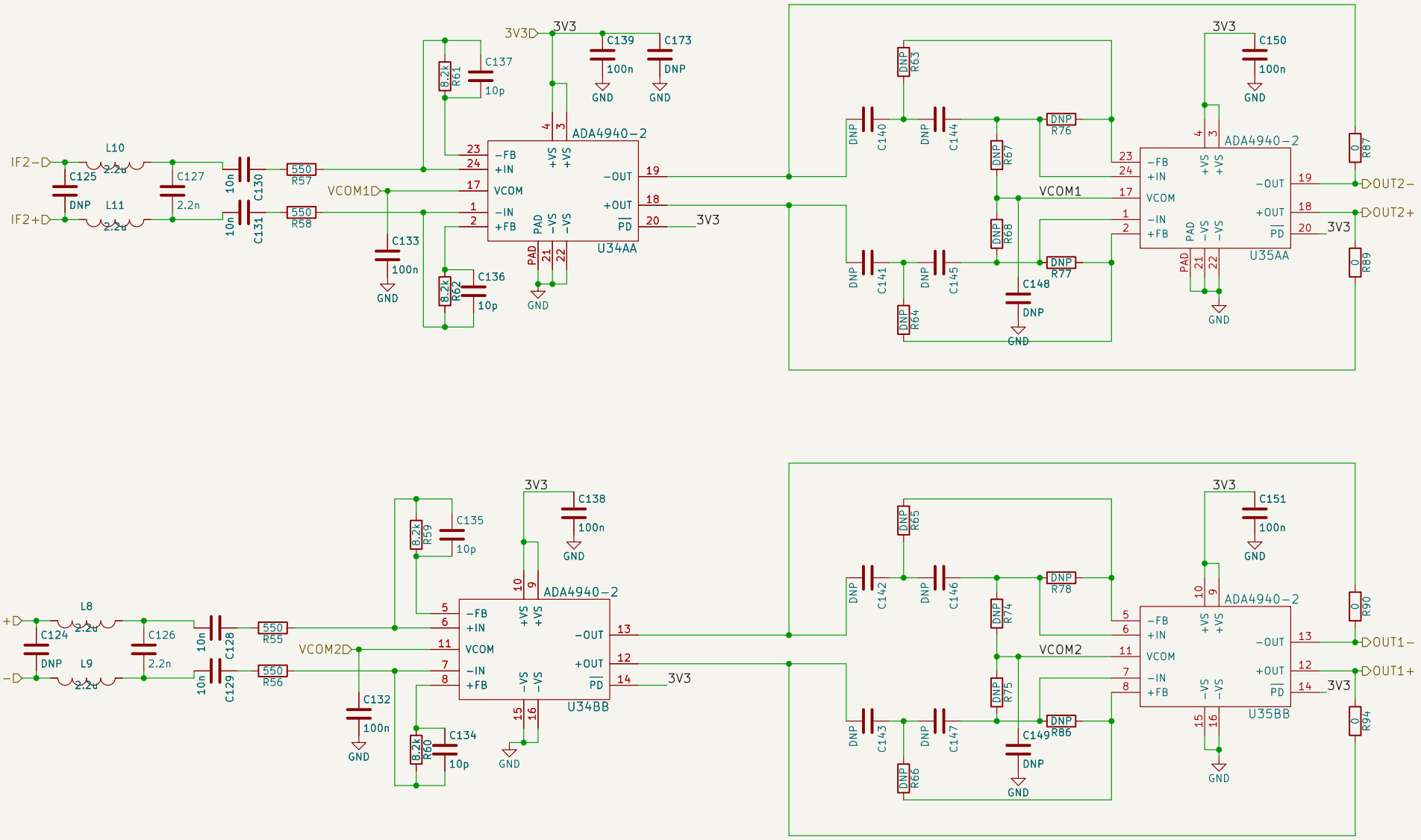


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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 4/10

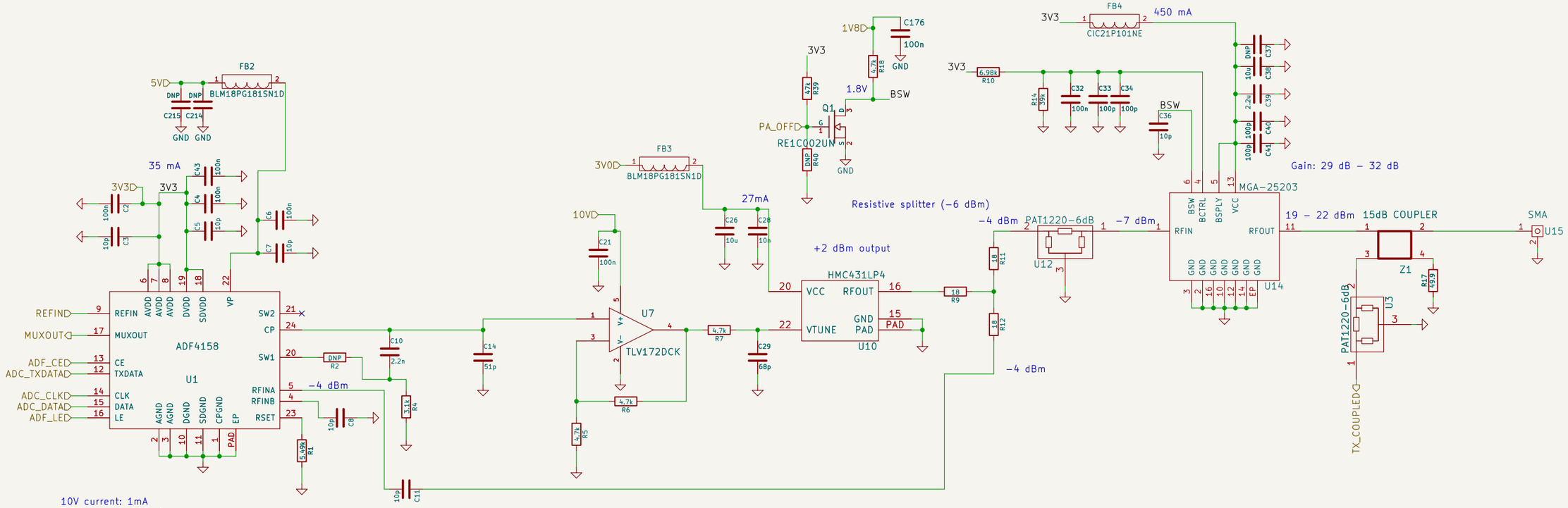

CON2
BARREL_JACK
FB11
CIC21P101NE
12V
GND
C64 10u
C70 10u
C71 DNP
C72 DNP
C177 DNP
C178 DNP
3V6
D1 LED
R24 240
L7980
U19
SYNC
L7
33u (SRR6040A-330M)
C62 10u
R28 59k
R30 8.2k
C69 68p
C66 10n
SS13FL D4
R36 4.99k
R37 590
C75 4.7n
R42 100
C79 10u
C82 10u
C83 10u
C84 10u
C169 DNP
5V6
L7980
U18
L6
33u (SRR6040A-330M)
C61 10u
R26 68k
R29 6.8k
C68 100p
C65 10n
SS13FL D3
R34 4.99k
R35 1k
C74 8.2n
R41 56
C77 10u
C80 10u
C81 DNP
3V6
L7980
U17
L5
33u (SRR6040A-330M)
R103 0
R101 DNP
R31 DNP
C60 10u
R25 68k
R27 15k
C67 39p
C63 10n
SS13FL D2
R32 4.99k
R33 7.5k
C73 15n
R38 33
C76 10u
C78 10u
1V0
TPS7A91
U24
C88 10u
C92 100n
C90 10m
R47 4.57k
R48 4.15k
C102 10u
C105 DNP
3V3A
U22
C86 10u
C90 100n
C96 10m
R43 10.5k
R44 2k
C100 10u
C103 DNP
5V
U23
C87 10u
C91 100n
C97 10m
R45 1.87k
R46 1.5k
C101 10u
C104 DNP
1V8
LP5907-3.0
U20
C85 10u
C93 10u
C172 DNP
3V0
MIC5301-3.3
U25
C94 10u
C106 10u
C175 DNP
3V3D
LP2985A-10DBVR
U21
C89 1u
C95 10n
C98 10u
10V
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 5/10

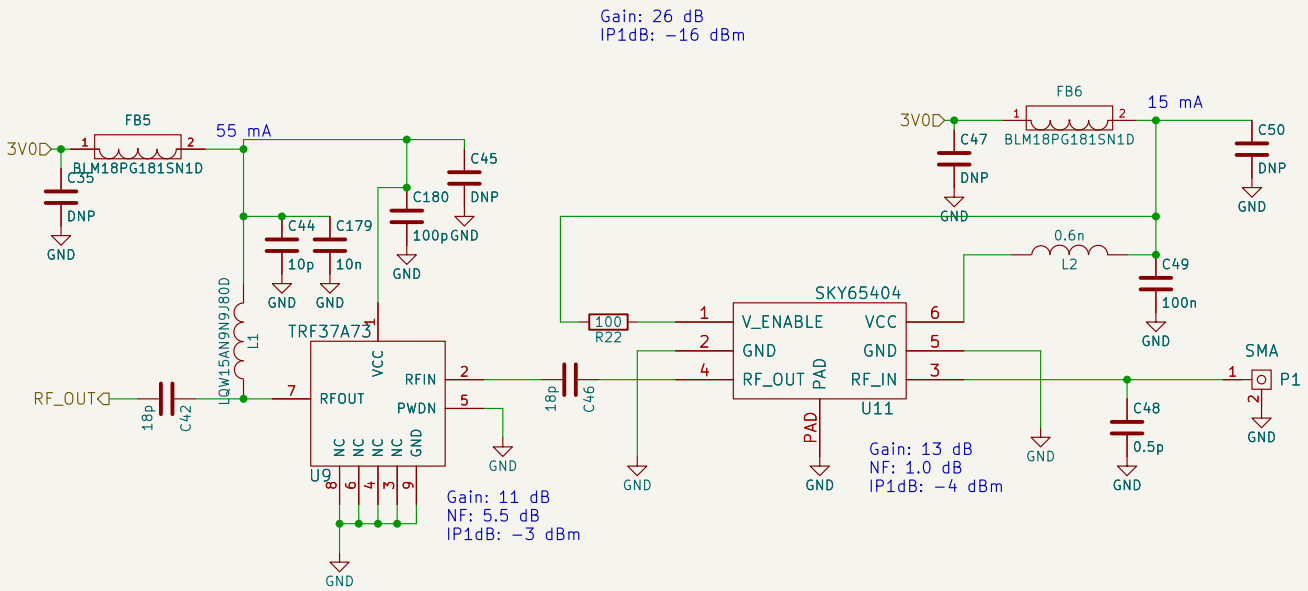


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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 6/10

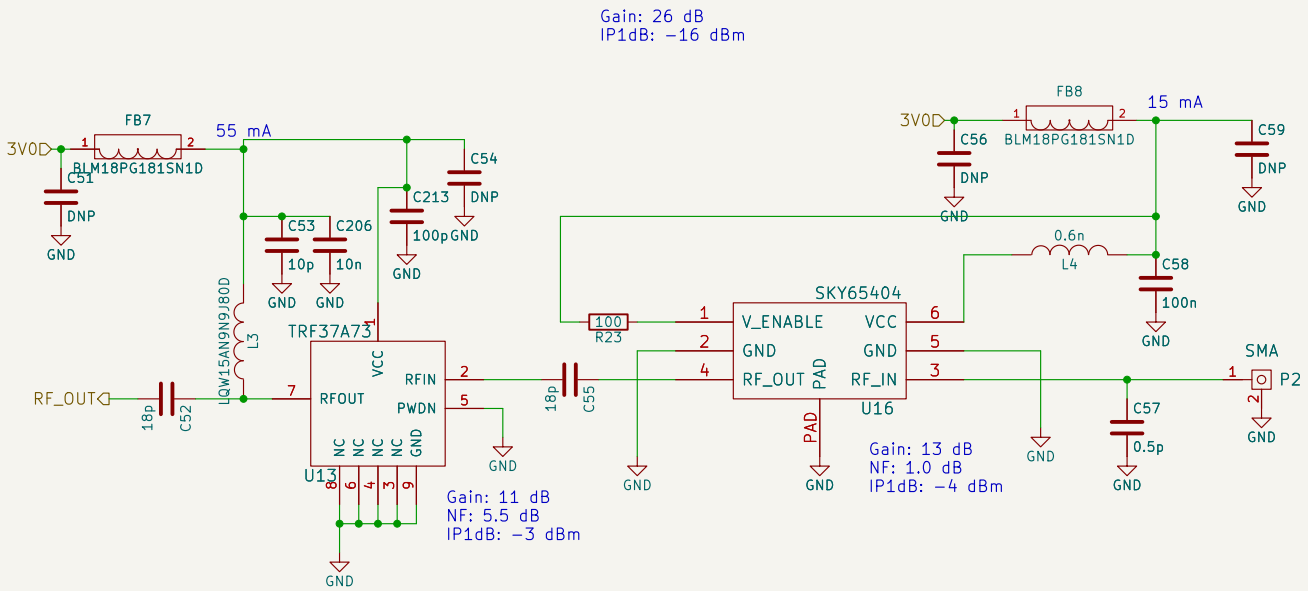


Sheet: /if/		
File: if.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 7/10


5V
FB2
BLM18PG181SN1D
DNP
DNP
C215
C214
GND GND
35 mA
3V3
3V3
C43 100n
C4 100n
C5 10p
C6 100n
C7 10p
100n C2
10p C3
AVDD AVDD AVDD DVDD SDVDD VP
REFIN
MUXOUT
ADF_CE
ADC_TXDATA
ADC_CLK
ADC_DATA
ADF_LE
REFIN
MUXOUT
CE
TXDATA
CLK
DATA
LE
ADF4158
U1
AGND AGND DGND SDGND CPGND EP
SW2
CP
SW1
RFINA
RFINB
RSET
DNP R2
C10 2.2n
C14 51p
5.49k R1
10p C8
−4 dBm
3.1k R4
10p C11
10V current: 1mA
3.3V current: 600mA
5V current: 1mA
3V0
FB3
BLM18PG181SN1D
27mA
C26 10u
C28 10n
10VD
C21 100n
U7
TLV172DCK
4.7k R7
4.7k R6
4.7k R5
HMC431LP4
VCC RFOUT
GND PAD
VTUNE
PAD
U10
C29 68p
+2 dBm output
18 R9
1V8
C176 100n
GND
4.7k R18
1.8V
BSW
47k R39
PA_OFFD
Q1
RE1C002UN
DNP R40
GND
Resistive splitter (−6 dBm)
−4 dBm
PAT1220−6dB
U12
−7 dBm
18 R11
18 R12
−4 dBm
3V3
FB4
CIC21P101NE
450 mA
3V3
6.98k R10
R14 39k
C32 100n
C33 100p
C34 100p
BSW
C36 10p
DNP C37
10u C38
22u C39
100p C40
100p C41
Gain: 29 dB − 32 dB
MGA−25203
BSW BCTRL BSPLY VCC
RFIN
RFOUT
GND GND GND GND GND GND GND GND EP
U14
19 − 22 dBm 15dB COUPLER
Z1
R17 49.9
SMA
U15
PAT1220−6dB
U3
TX_COUPLED
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Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.5
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Sheet: /rx1/		
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Size: A4	Date:	**Rev:**
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Sheet: /rx2/		
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 10/10